# SS2P2: A Stable State-Space Point Process for Limit-Order-Book Simulation

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## Abstract

Simulating limit-order-book (LOB) order flow is a core generative task for market-making world models, agent training, and counterfactual analysis. The state-space point process S2P2 predicts the next LOB event accurately, but we find that an uncapped decoder built on its backbone drifts or explodes when rolled out in closed loop. We propose SS2P2 (Stable State-Space Point Process), which keeps the backbone and replaces only the output heads: a softmin-bounded rate head whose intensity has a hard closed-form ceiling (an exact dominating rate for Ogata thinning), and a soft-max mark head that is rate-neutral at a fixed state. The bounded intensity makes SS2P2 stable in the point-process sense: the process is non-explosive, and the event count over any finite horizon is stochastically dominated by a homogeneous Poisson process at the ceiling. On three Coinbase assets, against five baselines with three training seeds each and empirically verified mean-rate calibration (15% tolerance), SS2P2 attains the best NLL and type accuracy on two of three assets and verifies calibration on all nine checkpoints, where the uncapped variant fails on six. SS2P2 trains with a parallel scan and simulates with constant-cost state updates by exact thinning: a likelihood-trained model that predicts accurately and simulates reliably within our protocol, a practical foundation for market-making world models.

## Introduction

The limit order book (LOB) is the core mechanism of electronic markets: a continuous-time stream of order placements, cancellations, and trades whose fine structure drives price discovery and liquidity. A generative model of this stream is asked two things. *Prediction*: given the event history, score the time and type of the next event. *Simulation*: roll the model forward freely, feeding its own samples back as history, and reproduce the stylized facts of real order flow — calibrated event rates, burst–lull clustering, heavy tails (Cont 2001). Simulation is the harder task and, for downstream use (market-making world models, agent training, counterfactual analysis), the more valuable one (Li et al. 2025; Nagy et al. 2025).

Neural MTPPs (Du et al. 2016; Mei and Eisner 2017; Shchur et al. 2021) excel at prediction, and the continuous-time state-space model S2P2 (Chang et al. 2025) reports state-of-the-art held-out likelihood across standard event-sequence benchmarks — but only teacher-forced: the S2P2 paper never rolls the model out. We evaluate a factorized, uncapped decoder built on the same backbone (S2P2$_{\mathrm{U}}$; see Background), and find it is not a usable simulator under our protocol. Two obstacles are responsible. *First, instability*: the self-excitation that wins the one-step likelihood becomes a liability in closed loop — each sampled event raises the intensity, which produces more events, and with no bound on the rate the roll-out drifts hot. *Second, failed calibration*: the standard patch, rescaling the intensity until the simulated rate matches reality, fails exactly where it is needed most. Trained S2P2$_{\mathrm{U}}$ checkpoints respond so sharply to the rescaling constant that our procedure finds no multiplier passing verification; on one of our three assets, every S2P2$_{\mathrm{U}}$ checkpoint fails. The gap is concrete: the strongest predictors in our benchmark are unusable or unreliable in closed loop, while simple stable models simulate safely but predict poorly.

SS2P2 changes only the output heads. The stacked latent-linear-Hawkes state-space backbone is shared verbatim with our S2P2$_{\mathrm{U}}$ baseline (diagonal zero-order-hold dynamics, parallel-scan computability, LayerNorm'd stack output $h_t$); everything after $h_t$ is replaced, so differences against the uncapped baseline are attributable to the heads. Throughout, *stable* means precisely this: the conditional intensity is globally bounded by an architectural constant, so the event count over any finite horizon is stochastically dominated by a homogeneous Poisson count and roll-outs cannot explode; the bound does not by itself guarantee stationarity, ergodicity, or realism, which are assessed empirically. Our contributions:

1. **The SS2P2 head pair**: a softmin-bounded rate head whose intensity has a hard closed-form ceiling (an exact dominating rate for Ogata thinning) and whose one-sided cap leaves the low-rate regime nearly untouched, times a soft-max mark head that cannot change the total rate at a fixed state. Together they make the simulator globally rate-bounded and finite-time non-explosive by construction.
2. **Stable training and simulation.** Stateful (TBPTT) training walks contiguous stretches of the stream so the model never trains in a cold-start transient, and closed-loop roll-outs carry the packed decoder state at constant cost per state update, mathematically equivalent to encoding the full history. Training and simulation see the same regime.

3. **Verified post-hoc mean-rate calibration.** The rate is exactly linear in a learnable scale, the thinning ceiling scales with it, and marks are untouched at a fixed state, so calibration is a one-dimensional root-find that preserves the architectural bound; a mark-ratio-preserving variant applies to every baseline, and every calibration must pass a full-scale verification roll-out within a stated tolerance. Under this empirical test the uncapped baseline frequently fails with high rescaling sensitivity, while the bounded head verified on all nine checkpoints.

## Background and Related Work

**Marked temporal point processes.** An MTPP models a sequence $\mathcal{H}_T = \{(t_i, e_i)\}_{i=1}^{N_T}$ of events with times $t_i$ and types $e_i \in \mathcal{E}$, a finite set of atomic event types, through per-type conditional intensities $\lambda_e(t \mid \mathcal{H}_t)$. With total intensity $\lambda(t) = \sum_e \lambda_e(t)$ and mark distribution $p(e \mid t) = \lambda_e(t)/\lambda(t)$, the log-likelihood is

$$\log \mathcal{L} = \sum_i \Big[\log \lambda(t_i) + \log p(e_i \mid t_i)\Big] - \int_0^T \lambda(u \mid \mathcal{H}_u)\, du,$$

the integral being the compensator $\Lambda(t) = \int_0^t \lambda(u \mid \mathcal{H}_u)\, du$. Sampling uses Ogata thinning (Lewis and Shedler 1979; Ogata 1981), which requires a dominating rate $\bar{\lambda} \geq \lambda(t)$; models exposing no practical dominating rate fall back on heuristic bound estimates, which can bias samples when the estimate is exceeded.

**The S2P2 backbone.** S2P2 (Chang et al. 2025) is a state-space point process: $L$ stacked latent-linear-Hawkes/SSM layers with diagonal state, ZOH-discretized between events and computable by parallel scan,

$$x_{t'}^{(\ell)} = \bar{A}^{(\ell)} x_t^{(\ell)} + (\bar{A}^{(\ell)} - I)\tilde{B}^{(\ell)} h^{(\ell-1)} \big[+\tilde{E}^{(\ell)}\alpha \text{ at an event}\big],$$

with a post-norm residual readout (LayerNorm (Ba, Kiros, and Hinton 2016) after each layer's residual sum) producing the continuous-time hidden representation $h_t := h^{(L)}(t) \in \mathbb{R}^H$. The published decoder reads *per-type* intensities from $h_t$ through a linear projection followed by a scaled softplus with a learnable per-type sharpness, $\lambda_e = \mathrm{softplus}\big(\beta_e\, a_e\big)/\beta_e$ with $a = W h_t + b$. Each $\beta_e$ reshapes the curvature near zero, but the map is asymptotically linear with unit slope, so the head carries no explicit output ceiling.

Our experiments use a *factorized, uncapped* variant, denoted $\mathrm{S2P2_U}$: a scalar total rate $\lambda = \mathrm{softplus}(w^\top h_t + b)$ on one half of $h_t$, a soft-max mark head on the other, built on our real-valued reimplementation of the backbone. The reimplementation keeps the published configuration's block structure (backward-ZOH input drive, input-dependent dynamics, post-norm GELU residual readout, event impulses, left-limit likelihood) and deviates in four respects: real diagonal states with log-spaced timescale initialization replace the complex DPLR spectrum; a learned skip projection replaces the identity residual with diagonal feedthrough; dynamics gates condition on each interval's end rather than its start; and between-event queries hold inter-layer inputs at their last-event values. $\mathrm{S2P2_U}$ shares the published head's operative property, a softplus rate with no explicit ceiling, but is not the published per-type decoder, and we label it accordingly. SS2P2 keeps everything up to and including $h_t$ and changes only the heads, so SS2P2–$\mathrm{S2P2_U}$ differences are attributable to the head pair.

**Neural TPPs for event streams.** Neural MTPPs learn history-dependent intensities with recurrent (Du et al. 2016; Mei and Eisner 2017), attention-based (Zhang et al. 2020; Zuo et al. 2020), jump-diffusion (Zhang et al. 2024), and continuous-time state-space (Chang et al. 2025) encoders (survey: Shchur et al. 2021). Shi and Cartlidge (2022) specialize neural Hawkes models to LOB streams and evaluate both prediction and simulation. The state-space line, however, is evaluated almost exclusively teacher-forced: Chang et al. (2025) report state-of-the-art held-out likelihood but never roll the model out.

**Hawkes modeling of order flow.** Parametric Hawkes processes are the classical LOB event model (Hawkes 1971, 2018; Jain et al. 2024a,b), with state-dependent extensions (Morariu-Patrichi and Pakkanen 2022): exact stationarity certificates, limited expressiveness. SS2P2 takes the complementary route: full neural expressiveness with a closed-form rate bound restoring non-explosiveness. In the neural TPP literature such bounds appear mainly as computational devices for thinning; here the bound is a modeling decision, one-sided so the quiet regime is left undistorted.

**Generative LOB simulators.** Learned order-flow generators span GANs (Li et al. 2020; Coletta et al. 2021, 2022), token-level autoregressive state-space models (Nagy et al. 2023), diffusion-guided agents (Huang et al. 2026), and foundation-model-scale transformers (Li et al. 2025; Kawawa-Beaudan et al. 2026). Agent-based simulators (Byrd, Hybinette, and Balch 2020; Frey et al. 2023) remain the RL workhorse. These systems are scored on distributional realism: stylized facts (Cont 2001; Vyetrenko et al. 2020) and the KS/Wasserstein suites of LOB-Bench (Nagy et al. 2025), whose authors report error accumulation over long roll-outs. SS2P2 differs on two counts: one likelihood-based model is held to *both* prediction and closed-loop realism, and the simulated mean rate is calibrated and checked by a full-scale verification roll-out, an empirical closed-loop test the simulators above do not report.

## Methods

To apply point process modeling to high-frequency LOB data, we represent market dynamics as a sequence of **events** (pipeline in Figure 1): we collect the data, construct events, and train SS2P2 to learn the underlying dynamics.

### Data collector

We obtain trade and Level-2 limit order book data from Kaiko (Kaiko 2025), an institutional provider of digital-asset market data. Each update record carries a timestamp, an update type (snapshot or liquidity update), and price–volume information per bid and ask level. Trades are retrieved via the same API and attributed to the book change whose update interval contains them, their volume netted against the

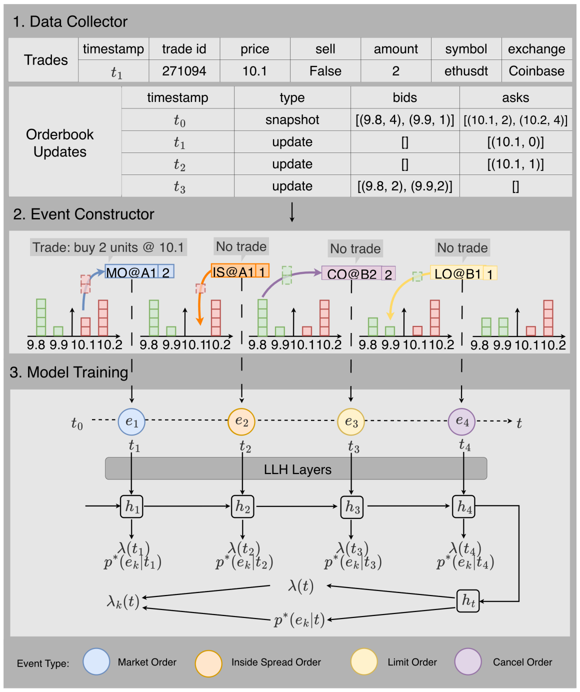


Figure 1: Pipeline. (1) *Data collector*: aligned trade records and Level-2 book updates. (2) *Event constructor*: consecutive snapshots are differenced and aligned with trades to emit atomic typed events. (3) *Model training*: the event sequence drives the state-space backbone whose hidden representation $h_t$ decodes the bounded total rate $\lambda(t)$ and mark distribution $p(e \mid t)$, giving $\lambda_e(t) = \lambda(t)\, p(e \mid t)$.

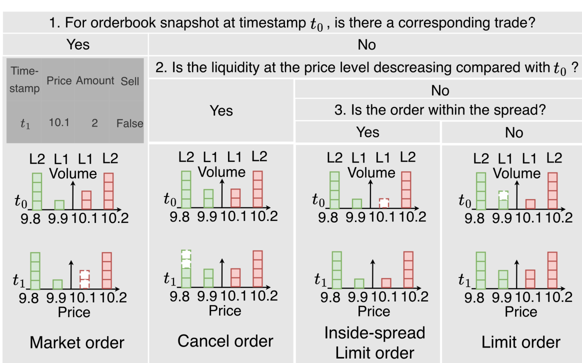


Figure 2: Event construction for the four order classes through three questions, each a decision-tree classification applied to synchronized Level-2 book snapshots and trade records at timestamps $t_0 \to t_1$.

liquidity change at the traded price; the event constructor classifies every observed book change through this alignment.

## Event constructor

We represent market dynamics as a sequence of **events** over the top $K = 10$ levels on each side.

**Atomic Events.** We define a finite set $\mathcal{E}$ of *atomic event types* $e = (c, s, \delta)$: a class $c \in \{\mathrm{LO}, \mathrm{MO}, \mathrm{CO}, \mathrm{IS}\}$ — limit order (liquidity provision at a top-$K$ level), market order (consumption at the best quote), cancel order (withdrawal at a top-$K$ level), or inside-spread order (a new order improving the prevailing best quote); a side $s \in \{b, a\}$; and a level index $\delta$. For LO/CO, $\delta \in \{1, \dots, K\}$ is the distance from the best quote ($\delta = 1$ = best bid/ask); for MO, $\delta = 1$ always; for IS, $\delta \in \{1, \dots, K\}$ is the number of ticks by which the new quote improves the spread.

**Event Construction from LOB and Trade Data.** As illustrated in Figure 2, atomic events are inferred by differencing consecutive Level-2 snapshots and aligning them with trades in the inter-update interval, via a three-step decision tree. A liquidity decrease with a matching trade at that price is a **Market Order** (recorded at $\delta = 1$; any residual decrease becomes a cancellation); a decrease with no matching trade is a **Cancel Order**; an increase is an **Inside-Spread Order** if inside the previous spread (indexed by tick distance from the previous best) and otherwise a **Limit Order** (indexed by its post-update level). One update can emit several events; they share its timestamp and are ordered deterministically (MO, CO, LO, IS; bid before ask; then by level). Order sizes are retained in the dataset but the models use only event times and types.

## Stable State-Space Point Process (SS2P2)

SS2P2 factors the per-type intensity into a bounded total rate and a mark distribution — the ground-intensity–mark factorization of set-valued MTPPs (Chang, Boyd, and Smyth 2024) — both decoded from the *same* continuous-time hidden representation $h_t$: $\lambda_e(t) = \lambda(t)\, p(e \mid t)$, with $\sum_e \lambda_e(t) = \lambda(t)$ since $\sum_e p(e \mid t) = 1$. The design principle is to put nonlinearity where it cannot perturb the rate.

## Softmin-bounded rate head

An uncapped head can fail in closed loop through feedback: each sampled event raises the intensity, faster events follow, and with nothing bounding the rate the roll-out drifts hot. The rate head is built to break this loop. From the backbone's hidden representation $h_t$ (LayerNorm'd, hence bounded in scale), a gate produces a bounded state $g$, read out through a smooth one-sided cap with fixed constant $z_{\max}$ ($z_{\max}$=6 in all experiments):

$$
\begin{aligned}
o &= \sigma(W_o h_t + b_o) \in (0,1)^H, \\
g &= o \odot \tanh(h_t) \in (-1,1)^H, \\
z_{\text{raw}} &= w^\top g + b, \\
z &= z_{\max} - \operatorname{softplus}\big(z_{\max} - z_{\text{raw}}\big), \\
\lambda(t) &= \varsigma \cdot \operatorname{softplus}(z)
\end{aligned}
$$

with a learnable positive scale $\varsigma$ initialized from the training-split rate ($b = 0$ at initialization, so $g \approx 0$ gives $\lambda \approx \varsigma \ln 2$ when $z_{\max} \gg 0$; appendix).

- **Hard ceiling.** Since $\operatorname{softplus}(x) > 0$ for every $x$, $z \leq z_{\max}$ holds identically, so for every state and parameter setting $\lambda(t) \leq \bar{\lambda} := \varsigma \operatorname{softplus}(z_{\max})$. The intensity is globally bounded, the event count on any finite horizon $[0, T]$ is stochastically dominated by a homogeneous Poisson count with mean $\bar{\lambda}T$ (finite-time non-explosion), and $\bar{\lambda}$ is an *exact* dominating rate for Ogata thinning (Proposition 1, appendix). The bound guarantees no more: stationarity, ergodicity, and realism are assessed empirically.

- **Bounded input.** The readout sees $g \in (-1, 1)^H$: however far the closed loop drives $h_t$, an extreme state changes the direction of $g$ but not its magnitude. For fixed parameters this already gives $|z_{\text{raw}} - b| < \|w\|_1$, a computable parameter-dependent ceiling even without the cap; the cap adds a ceiling that is *user-chosen, independent of* $w$ *and* $b$*, and invariant under training*.
- **Vanishing gain.** Differentiating the cap gives $\partial z / \partial z_{\text{raw}} = \sigma(z_{\max} - z_{\text{raw}})$: as a burst pushes the intensity toward the ceiling, the head becomes exponentially *less* sensitive to further state excursions, so the feedback loop loses gain exactly when it is most dangerous. In our experiments this coincided with a closed-loop rate that responds smoothly to rescaling (see Calibratability).
- **One-sided cap.** The cap distorts only the high end: for $z_{\text{raw}}$ well below $z_{\max}$, $z \approx z_{\text{raw}}$ and the log-intensity derivative approaches one, so quiet-period gradients pass essentially unattenuated. Because $g$ is bounded, the intensity has a strictly positive, parameter-dependent floor (typically numerically negligible; Proposition 2, appendix) rather than an exact zero.

The first three choices are intended to control the intensity ceiling and its sensitivity; the fourth keeps the bound from distorting prediction when the market is quiet.

## Rate-neutral mark head

Marks are decoded from the same $h_t$ and are *instantaneously rate-neutral at a fixed hidden state*:

$$p(e \mid t) = \operatorname{softmax}\big(\operatorname{MLP}(h_t)\big)_e, \qquad \lambda_e(t) = \lambda(t)\, p(e \mid t),$$

so $\sum_e \lambda_e(t) = \lambda(t)$ exactly, whatever the mark logits. Because the soft-max lives on the simplex, the mark network can be arbitrarily deep, or later conditioned on exogenous state such as an agent's resting quotes, without changing the total rate at that state or the architectural ceiling. Two qualifications: in closed loop, a changed mark distribution changes sampled marks, hence future states and rates (the ceiling holds, but the realized mean rate can shift and may require recalibration); and both heads read the shared backbone, so mark gradients still shape $h_t$ — only the head-specific parameters separate. The simplex guarantees rate neutrality state by state, not trajectory by trajectory.

## Likelihood and sampling

Training minimizes the negative log-likelihood of each training window, covering its events $(t_i, e_i)$ and the next event after the window:

$$\mathcal{L} = \int_0^T \lambda(u \mid \mathcal{H}_u)\, du \;-\; \sum_i \Big[\log \lambda(t_i) + \log p(e_i \mid t_i)\Big],$$

with $T$ the next-event time. The integral and the $\log \lambda$ terms score *when* events happen (the compensator charges for intensity placed where no event occurred); the $\log p$ term is a categorical cross-entropy scoring *what* each event is. By the factorization, the timing terms are the only loss on the rate head's parameters and the mark term the only loss on the mark head's; the shared backbone receives both gradients. Each $\lambda(t_i)$ is evaluated at the left limit, so an event never informs its own intensity. During *training*, the compensator is approximated by one intensity evaluation per inter-event interval (the endpoint rule), which under-weights a decaying intensity and biases the learned rate upward; the post-hoc calibration below compensates the resulting inflation of the realized closed-loop mean rate, not the likelihood approximation itself. Held-out NLL is evaluated with a denser 32-point log-spaced quadrature per gap. Closed-loop sampling uses Ogata thinning with the exact dominating rate $\bar{\lambda} = \varsigma \cdot \operatorname{softplus}(z_{\max})$: propose arrivals from a homogeneous Poisson process at the ceiling and accept each with probability $\lambda(t)/\bar{\lambda}$.

## Carried-state simulation

The standard roll-out protocol re-encodes a sliding window of the last $W$ events from a zero state at every step: $O(W)$ work per simulated event and a hard memory ceiling — nothing older than $W$ events can influence the future, so long-memory stylized facts are *architecturally* unreachable regardless of the model. Because the backbone is a state-space recursion, both costs vanish: we carry the packed decoder state and advance it with only the new event, mathematically equivalent to encoding the entire history from the seed. The state update and each thinning proposal cost $O(1)$ in the history length (expected proposals per accepted event = $\bar{\lambda}/\lambda$), removing the $O(W)$ re-encode and the memory ceiling. Attention- and window-shaped baselines have no such Markov state and retain the sliding window.

## Stateful (TBPTT) training

Windowed maximum likelihood encodes every training window from a cold initial state, so the model only ever trains in a "history just began" transient and MLE explains the observed rate with an inflated baseline — the free roll-out, whose state is warm, then double-counts. We train instead with truncated backpropagation through time (Williams and Peng 1990): batch lanes walk contiguous stretches of the stream in order, and the decoder state at the end of window $t$ is passed, *detached*, as the initial state of window $t+1$ (resets only at genuine stream boundaries). Gradients remain window-local; state values flow from the start of the stream. Training is thus consistent with carried-state simulation at unchanged per-step cost; it requires non-overlapping windows (stride = window length), and our controls match that setting.

## Post-hoc rate calibration

The factorization buys one more property: $\lambda = \varsigma \cdot \operatorname{softplus}(z)$ is *exactly linear* in the learnable scale $\varsigma$, the thinning ceiling $\varsigma \cdot \operatorname{softplus}(z_{\max})$ scales with it, and mark probabilities are untouched by construction. The free-running mean rate is nonlinear in $\varsigma$ through the closed-loop feedback; we do not prove it monotone, but it was monotone over every successful bracket in our experiments. We therefore calibrate by a one-dimensional geometric bisection on a multiplier $\kappa$: probe roll-outs at $\kappa\varsigma$ bracket the target, then bisect in log space (full protocol in the appendix). The result, when it verifies, is a simulator whose thinning bound stays exact (the ceiling scales with $\kappa\varsigma$) and whose realized mean rate matched the target within tolerance at the rates we examined; the calibration is empirical, not exact. For *any* decoder, a constant $\kappa$ multiplying the sim-time intensity preserves conditional type ratios $\lambda_e/\lambda$ at a fixed history (marginal mark frequencies can still shift in a new roll-out, since histories change), so the same bisection applies to every baseline; what is lost off-family is only the exactness of the thinning bound. The procedure is falsifiable (targets, tolerances, and verification in Calibratability and the appendix); unresolved failures exclude the checkpoint and are reported as results, and transfer to the test split is reported, never gated. We use calibration as a *post-hoc* correction for the realized-rate inflation caused by the biased endpoint-rule training compensator — and, in the experiments, as a measurement instrument in its own right.

## Experiments and Evaluations

We evaluate SS2P2 against five baselines on three questions: (1) does the bounded head reduce predictive accuracy? (2) does SS2P2 reproduce real order-flow structure in closed loop? (3) which models pass verified rate calibration? We close with efficiency.

### Experimental Setup

**Datasets.** We benchmark on Coinbase BTC, ETH, and SOL event-driven order flow, seven days per asset, encoded into $|\mathcal{E}| = 62$ atomic types $(20 + 20 + 2 + 20$ over LO/CO/MO/IS, sides, and levels). Validation-split rates are 38.3, 47.9, and 24.0 events/s: production rates, not subsampled streams.

**Baselines.** All models share the same input embedding and training configuration. They differ only in the decoder.

- **LSTM**: a standard LSTM decoder.
- **NHP** (Mei and Eisner 2017): the neural Hawkes continuous-time LSTM.
- **SAHP** (Zhang et al. 2020): an SAHP-style adaptation, not a faithful reproduction: a two-layer, four-head causal transformer encoder with a softplus intensity held piecewise-constant between events (the published decay parameterization is not ported).
- **PCT-LSTM** (Shi and Cartlidge 2022): the state-dependent parallel neural Hawkes process (CT4LSTM-PPP) *without its exogenous market-state input*, which the closed-loop protocol cannot supply (this may cost it accuracy); the event-driven cell is ported unchanged.
- **S2P2$_\mathrm{U}$**: our factorized, uncapped ablation of the S2P2 decoder (Chang et al. 2025): identical backbone to SS2P2, scalar softplus rate with no ceiling (Background; not the published per-type head).

**Training and evaluation protocol.** Each model trains per asset with sequence length 1024, non-overlapping windows, identical gradient-step counts, and three training seeds. The fixed 1024-event context spans only $\sim$20–40 s of market time at these rates, so long-memory structure leans entirely on carried state. Evaluation is strict end-to-end:

- **Prediction** is scored on *every* genuine test event by a lossless streaming evaluator with carried state (one cold start per genuine segment boundary); it hard-fails if any event goes unscored. NLL uses a 32-point log-spaced quadrature per gap; type accuracy is conditional on the observed next-event time; MAE is expected-time absolute error on a 60 s horizon (definitions in the appendix).
- **Simulation** uses carried-state closed-loop roll-outs ($32 \times$ 600 s, three roll-out seeds per checkpoint) against equal-duration bootstrap samples of real segments. SS2P2 samples by Ogata thinning with its exact closed-form ceiling; the baseline heads expose no closed-form dominating rate and share a common compensator-inversion sampler. Stylized facts are computed on an order-flow proxy (signed counts of price-moving events per 1 s bucket), identically for real and simulated streams (appendix).
- **Calibration**: models are rate-calibrated before scoring by a one-dimensional bisection on the time-rescaling constant $\kappa$, targeting the *validation*-split rate; a full-scale validation roll-out must then reproduce the target within 15% (protocol in the appendix). Failing checkpoints are excluded and the failure is reported as a result. SAHP is the exception: after its free-running rate collapsed on every asset, calibration was not attempted for it (protocol decision) and it is reported uncalibrated (†).
- **Statistics**: roll-out seeds are averaged within each checkpoint. 95% CIs are $t$-based across three training seeds, indicative only at this sample size. Any batch failure or non-finite value aborts the run. Released scripts regenerate every table and figure.

### Prediction Quality

SS2P2 leads NLL and type accuracy on two of the three assets, and bounding the intensity did not reduce predictive performance in these experiments. In Table 1, SS2P2 wins overall NLL on SOL by a wide margin (0.27 vs. the strongest baseline's 0.38) with the best accuracy (0.192 vs. 0.184), wins both outright on ETH ($-0.87$ vs. PCT-LSTM's $-0.84$; 0.345 vs. NHP's 0.341), and on BTC ties NHP on marks (0.448 vs. 0.449) while conceding overall likelihood. Expected-time MAE is within 0.030–0.050 s of the truth on every asset, approximately 1–3 ms behind the best model per asset (PCT-LSTM has the lowest MAE on ETH and SOL); the uncapped S2P2$_\mathrm{U}$ is pathological (2.3, 10.2, 46.2 s), consistent with near-zero-intensity states whose expected-time integral runs to the evaluation horizon.

### Simulation Quality

SS2P2 is the only model pairing reliable closed-loop simulation with top-tier prediction; the two strongest baseline predictors are unusable or erratic in closed loop. SAHP's

| model | BTC | | | ETH | | | SOL | | |
|---|---|---|---|---|---|---|---|---|---|
| | NLL↓ | ACC↑ | MAE↓ | NLL↓ | ACC↑ | MAE↓ | NLL↓ | ACC↑ | MAE↓ |
| NHP | **-1.07±0.01** | **0.449±0.003** | **0.029±0.0001** | -0.81±0.03 | 0.341±0.006 | 0.030±0.0003 | 0.53±0.01 | 0.184±0.004 | 0.052±0.001 |
| LSTM | -0.44±0.40 | 0.411±0.015 | 0.032±0.002 | 0.15±0.04 | 0.288±0.003 | 0.034±0.002 | 1.34±0.25 | 0.141±0.005 | 0.057±0.004 |
| SAHP † | -0.93±0.01 | 0.412±0.001 | 0.034±0.005 | -0.60±0.01 | 0.297±0.004 | 0.038±0.006 | 0.67±0.02 | 0.155±0.003 | 0.060±0.002 |
| PCT-LSTM | -0.97±0.03 | 0.380±0.009 | 0.029±0.0004 | -0.84±0.05 | 0.272±0.014 | **0.028±0.0002** | 0.38±0.08 | 0.130±0.011 | **0.048±0.0004** |
| $\text{S2P2}_{\text{U}}$ | -0.39±0.25 | 0.439±0.002 | 2.3±5.9 | -0.38±0.22 | 0.331±0.008 | 10.2±31.0 | 0.67±0.49 | 0.182±0.002 | 46.2±96.0 |
| SS2P2 | -0.87±0.12 | 0.448±0.001 | 0.030±0.001 | **-0.87±0.08** | **0.345±0.003** | 0.031±0.001 | **0.27±0.05** | **0.192±0.001** | 0.050±0.001 |

Table 1: Prediction per genuine test event (mean ± 95% CI over three training seeds; streaming lossless evaluator). NLL = overall negative log-likelihood (time + mark); ACC = next-type accuracy at the observed event time; MAE = expected-time mean absolute error in seconds (60 s evaluation horizon). SS2P2 wins NLL and ACC on ETH and SOL and ties NHP on BTC accuracy; $\text{S2P2}_{\text{U}}$'s MAE column reflects expected times that run into the evaluator's horizon clamp on a subset of states.

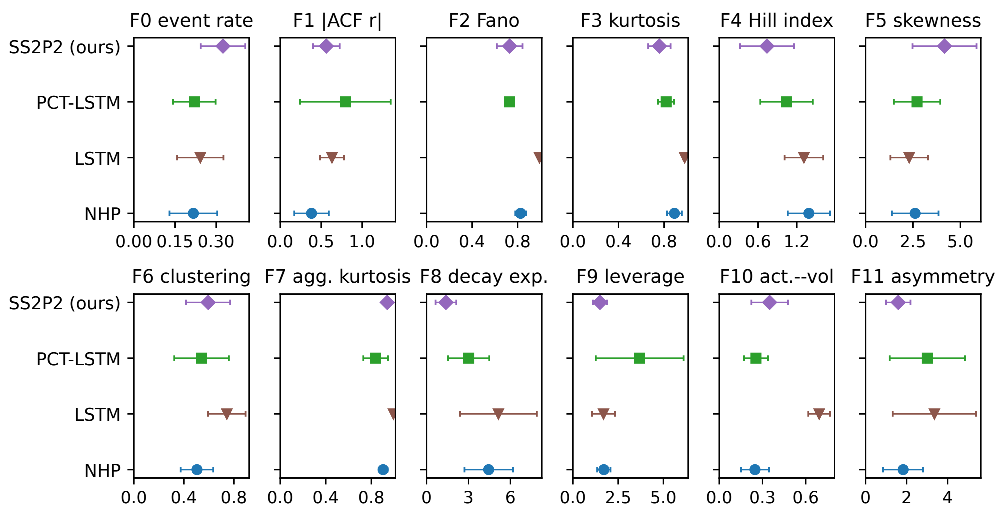


Figure 3: The stylized-fact battery on the order-flow proxy (identical construction for real and simulated streams), pooled across assets: relative error vs. real for facts F0–F11 (whiskers = 95% $t$-CI over asset–checkpoint pairs, indicative at $n$=3 seeds; roll-out seeds averaged within checkpoint). $\text{S2P2}_{\text{U}}$ and SAHP omitted: too few calibrated checkpoints.

free-running rate collapses to 5–16 ev/s against matched real segments at 20–35 ev/s, under half the real rate on every asset; per protocol it was not calibrated († in Table 1). The uncapped $\text{S2P2}_{\text{U}}$ drifts hot instead: 6 of 9 checkpoints, all three on SOL among them, fail verified calibration (Table 2), and the three that pass are erratic, with roll-out Fano relative errors of 0.27, 2.2, and 6.0: simulated dispersion quality is seed luck.

Among the remaining models, Figure 3 scores the roll-outs on the stylized-fact battery (twelve facts on the order-flow proxy), pooled across assets. SS2P2 has the lowest mean relative error on five of twelve (heavy tails F3, Hill index F4, volatility decay F8, leverage F9, asymmetry F11), ties PCT-LSTM for best on the Fano panel F2 (0.73 both), and is mid-pack on the rest; confidence intervals overlap heavily on most panels, so we read the figure as broad competitiveness, not dominance. The F2 panel still separates the families most sharply: the LSTM pins at ≈0.99 with a negligible interval because it simulates a near-Poisson stream, the trivially-stable end of the trade-off, while real Fano rises from ∼40–70 at 1 s to ∼340–1400 at 50 s across assets. NHP, the best BTC predictor, leads the rate and activity panels (F0, F10) but is mid-pack on burst structure; PCT-LSTM matches SS2P2 on F2 and leads aggregated kurtosis F7, but has the weakest type accuracy in Table 1. The reading we take is that prediction and simulation quality are distinct axes: in this benchmark no baseline wins both, and only the bounded design tracks burst structure while also predicting at the top of the table.

## Calibratability

Post-hoc rate calibration applies to *every* decoder, because a constant $\kappa$ rescales sim-time intensity while preserving conditional type ratios at a fixed history; we calibrate every model except SAHP (excluded above) and treat the *outcome* as a result. The protocol is falsifiable: probes must bracket and converge to the validation target, a full-scale roll-out must reproduce it within 15%, and every failure is reproducible (deterministic probe seeds; retried failures reproduce their brackets bit-identically).

| model | BTC 38 ev/s | ETH 48 ev/s | SOL 24 ev/s |
|---|---|---|---|
| NHP | 3/3 | 3/3 | 3/3 |
| LSTM | 3/3 | 3/3 | 3/3 |
| SAHP † | *rate collapse; not calibrated (protocol)* | | |
| PCT-LSTM | 3/3 | 3/3 | 3/3 |
| $\mathrm{S2P2_U}$ | **2/3** | **1/3** | **0/3** |
| SS2P2 | 3/3 | 3/3 | 3/3 |

Table 2: Checkpoints passing calibration + full-scale verification, out of three training seeds per asset (bold marks failures). The uncapped $\mathrm{S2P2_U}$ head fails on 6 of 9 checkpoints, including *all three* on SOL where no usable simulator remains, while the bounded SS2P2 head verifies on all nine. SAHP's free-running rate collapses on every asset; it was not calibrated (protocol decision) and is reported uncalibrated.

Whether a checkpoint passes this test is measurable, and in our benchmark the bound decided it: Table 2 shows the bounded SS2P2 head verifying on *all nine* checkpoints, while the uncapped $\mathrm{S2P2_U}$ head fails on 6 of 9, and on SOL the model class is excluded outright.

Figure 4 shows the failure in detail (shaded band: the bisection's $\pm 5\%$ probe tolerance; the 15% rule applies to full-scale verification). The failing BTC $\mathrm{S2P2_U}$ checkpoint responds extremely sharply to $\kappa$: an $8 \times 10^{-4}$ change moves the probe rate from below the band to above it ($34.8 \rightarrow 42.3$ ev/s against a $38.3$ target), repeat probes at the same $\kappa$ differ by tens of percent (in places nearly $2\times$), and the full-scale verification then missed by 22%. This high rescaling sensitivity is consistent with near-critical behavior; within the probed brackets the procedure found no multiplier satisfying verification, while the SS2P2 response crossed the same target smoothly on every asset. The same uncapped head whose near-zero-intensity states produce $\mathrm{S2P2_U}$'s pathological expected-time tail (Table 1) also produced the erratic closed-loop rate response, and both problems were absent under the softmin bound in every run we report — a strong empirical regularity, not a theorem.

## Efficiency Analysis

The state-space backbone is also computationally advantageous once its per-event sequential loop is replaced by a Hillis–Steele prefix scan (Hillis and Steele 1986) over the ZOH recurrence (equivalent up to floating-point evaluation order; maximum observed state/gradient discrepancy $2 \times 10^{-4}$ in our checks). On one GPU (Table 3, appendix; ETH checkpoints) the scan collapses the S2P2-family training step from $3.5$ s to $\sim 0.06$ s: attention-class speed, $60\times$ faster than the loop and $73\times$ faster than the CT-LSTM. For sampling, SS2P2's exact ceiling makes Ogata thinning available with a *constant* dominating rate (mean acceptance $0.54$ on ETH, $\sim 1.9$ proposals per accepted event): exactness costs $\sim 2.5\times$ throughput against the approximate inversion sampler shared by all models, and both run far above real time (353 vs. $48$ events/s on the fastest asset).

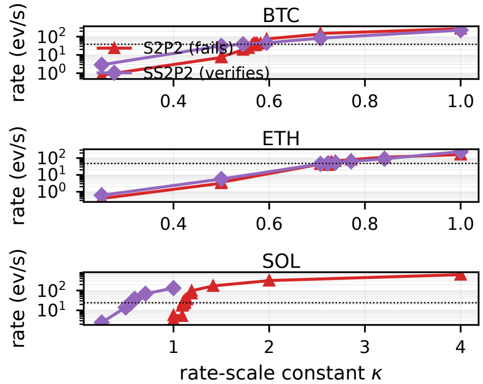


Figure 4: Failing $\mathrm{S2P2_U}$ checkpoints respond near-discontinuously to the rescaling constant $\kappa$ on every asset, most extremely on SOL; verified SS2P2 checkpoints cross each target smoothly. Probe ladders, seed-1 checkpoints; shaded band = $\pm 5\%$ probe tolerance around the validation target.

## Discussion and Conclusion

We held six neural MTPPs to both sides of the prediction–simulation tension on three Coinbase assets at production rates (24–48 events/s), under a protocol where failures count as results. The uncapped decoder $\mathrm{S2P2_U}$ is not a reliable simulator: it fails verified calibration on 6 of 9 checkpoints, including all three on SOL, and the three that pass are erratic across seeds. SS2P2, differing only in the output heads, achieves the best SOL NLL and type accuracy (PCT-LSTM has slightly lower MAE there), leads on ETH, ties the strongest baseline on BTC marks, verifies calibration on all nine checkpoints, and has the lowest error on five of twelve stylized-fact metrics while competitive on the rest.

First, then, closed-loop reliability tracked the intensity head, not training quality: the uncapped head's failures are reproducible, with high rescaling sensitivity consistent with near-critical behavior, while the bounded head's calibration reduced to a smooth root-find that verified every time. Second, the bound was nearly free: it did not reduce predictive performance in the reported experiments, trains at scan speed, and buys exact Ogata thinning at throughput far above real time.

The evidence has limits: three streams from one crypto venue, seven days each, and the simulator emits event types and times, so realism is scored on signed order-flow statistics, not asset price paths. The guarantee is a bounded rate, not realism: the LSTM is stable yet near-Poisson, even SS2P2 leaves a Fano dispersion gap, BTC timing headroom remains (NHP leads), and component-level attribution (gate vs. cap vs. factorization) awaits ablations. Because the mark head is instantaneously rate-neutral at a fixed state, conditioning it on an agent's resting quotes cannot violate the ceiling (recalibration may still be needed), making SS2P2 a candidate market-making world-model environment; this and the fill-probability queries of Chang, Boyd, and Smyth (2024) are future work.

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# Technical Appendix

## A. The architectural rate bound

**Proposition 1 (global ceiling).** For the head $g = o \odot \tanh(h_t)$, $z_{\text{raw}} = w^\top g + b$, $z = z_{\max} - \text{softplus}(z_{\max} - z_{\text{raw}})$, $\lambda = \varsigma\,\text{softplus}(z)$ with $\varsigma > 0$:

$$\lambda(t) \;\le\; \bar{\lambda} := \varsigma\,\text{softplus}(z_{\max})$$

for every $h_t$ and every $(W_o, b_o, w, b)$. *Proof.* $\text{softplus}(x) > 0$ for all $x$, so $z = z_{\max} - \text{softplus}(z_{\max} - z_{\text{raw}}) < z_{\max}$; softplus is increasing, so $\lambda < \varsigma\,\text{softplus}(z_{\max})$. $\square$

**Consequences.** (i) The conditional intensity is globally bounded, so for any finite horizon $T$ the compensator satisfies $\Lambda(T) \le \bar{\lambda}T < \infty$ and the process is non-explosive on $[0,T]$. (ii) The event count $N(T)$ is stochastically dominated by a homogeneous Poisson count with mean $\bar{\lambda}T$: the process can be constructed by thinning a homogeneous Poisson process of rate $\bar{\lambda}$, and thinning only removes points. (iii) $\bar{\lambda}$ is an exact dominating rate for Ogata thinning (Lewis and Shedler 1979; Ogata 1981). The bound establishes nothing beyond (i)–(iii); stationarity, ergodicity, and realism are empirical matters.

**Proposition 2 (parameter-dependent floor and pre-cap bound).** Since $g \in (-1,1)^H$, for fixed trained parameters $b - \|w\|_1 < z_{\text{raw}} < b + \|w\|_1$. Hence, even without the cap, $\lambda \le \varsigma\,\text{softplus}(b + \|w\|_1)$ is a computable post-training ceiling; and with the cap,

$$\lambda \;\ge\; \varsigma\,\text{softplus}\big(z_{\max} - \text{softplus}(z_{\max} - b + \|w\|_1)\big) \;>\; 0,$$

a strictly positive, typically numerically negligible floor. The intensity therefore has zero infimum only in the limit of unbounded parameters, not for any fixed trained model. The cap's contribution over the pre-cap bound is that $\bar{\lambda}$ is user-chosen, independent of the trained magnitudes of $w$ and $b$, and invariant under training.

## B. Calibration protocol

Targets are validation-split rates (38.27, 47.92, 23.98 ev/s for BTC, ETH, SOL). For each checkpoint: (1) probe roll-outs of 600 s at multiplier $\kappa$ starting from a geometric grid until the target is bracketed; (2) bisection in $\log\kappa$, at most ten steps, until a probe lands within 5% of the target; (3) a full-scale validation roll-out must reproduce the target within 15%; (4) failures escalate to full-scale probes, and unresolved failures exclude the checkpoint and are reported. Probe seeds are deterministic; retried failures reproduce their bisection brackets bit-identically. Monotonicity of the closed-loop rate in $\kappa$ is not assumed or proven; it held empirically over every successful bracket. Calibrated multipliers $\kappa$ (seeds 1/2/3): SS2P2 BTC 0.545/0.569/0.648, ETH 0.723/0.707/0.582, SOL 0.557/0.595/0.621; $\text{S2P2}_{\text{U}}$ (passing checkpoints only) BTC 0.534 (s2), 0.751 (s3), ETH 1.102 (s3).

## C. Metric definitions

**NLL** is per-event negative log-likelihood with the compensator integrated by a 32-point log-spaced quadrature per inter-event gap (horizon 60 s; beyond the grid the intensity is extended at its terminal value). **Type accuracy** is top-1 accuracy of $p(e \mid t^*)$ evaluated at the *observed* next-event time $t^*$ (accuracy conditional on the observed time). **Expected-time MAE** computes $\hat{\mathbb{E}}[\Delta t]$ by survival quadrature on the same log-spaced grid, adds the tail term $S(T_{\max})/\lambda(T_{\max})$, clamps the result to $[10^{-6}, 2T_{\max}]$ with $T_{\max} = 60$ s, and reports the mean absolute error against the true gap. All prediction metrics are computed by a streaming evaluator with carried state, one cold start per genuine stream segment, which hard-fails if any test event goes unscored.

## D. Stylized-fact construction (order-flow proxy)

The models emit event times and types only, so the battery operates on an order-flow proxy applied identically to real and simulated streams. Events are bucketed at 1 s. Per bucket, *activity* $a_t$ is the total event count and the *proxy return* $r_t$ is the signed count of price-moving events (unit tick): market orders and best-level inside-spread improvements, bid side positive, ask side negative. F0 is the event rate; F2 is the Fano factor $\text{Var}/\text{Mean}$ of bucket counts aggregated at scales $\{1, 2, 5, 10, 20, 50\}$ s, averaged over scales; F1, F3–F9, and F11 are standard stylized-fact statistics of $r_t$ (autocorrelation, kurtosis, Hill index, skewness, clustering, aggregated kurtosis, power-law decay of $|r|$ autocorrelation, leverage, timescale asymmetry); F10 correlates $a_t$ with $|r_t|$ (activity–volatility). None of these are statistics of asset price returns; we label them accordingly.

## E. Model and training configuration

All models share the input embedding (channel embedding 64, time embedding 64) and differ only in the decoder. Backbone/decoder hidden size 64, two layers. Training: AdamW, learning rate $2 \times 10^{-3}$, weight decay $10^{-6}$, gradient clipping at norm 0.5, ReduceLROnPlateau (factor 0.5, patience 5), batch 64 lanes, window 1024 events with

| | train step (s) | vs. seq. loop | sim. ev/s |
|---|---|---|---|
| NHP | 4.35 | – | 1253 |
| LSTM | 0.019 | – | 2083 |
| SAHP | 0.059 | – | 583 |
| PCT-LSTM | 0.79 | – | 1919 |
| S2P2$_{\mathrm{U}}$ | 0.055 | 64× | 1163 |
| SS2P2 | 0.059 | 60× | 879 / 353$^{\dagger}$ |

Table 3: Efficiency on one GPU. Training: one fwd+bwd on $64 \times 1024$ events (scan for the S2P2 family; speedup vs. their sequential loop). Simulation: events per wall-second, $8 \times 60\,\mathrm{s}$ roll-outs. $^{\dagger}$inversion / exact-ceiling Ogata thinning.

stride 1024 (non-overlapping, TBPTT with detached carried state), 12 epochs, best checkpoint by validation loss, three training seeds $\{1, 2, 3\}$, TF32 matmuls, no dropout. SS2P2 rate head: $z_{\max} = 6$; $\varsigma$ log-parameterized and initialized from the training-split rate ($\varsigma_0 = \text{rate}/\ln 2$); gate bias and rate bias initialized to zero; rate weights Xavier-uniform (gain $0.5$). Simulation: carried-state roll-outs, 32 sequences $\times$ $600\,\mathrm{s}$, three roll-out seeds, thinning for SS2P2 (constant ceiling $\varsigma\,\mathrm{softplus}(z_{\max})$), compensator-inversion sampling for baselines.

## F. Efficiency table

## G. Dataset

Coinbase BTC-USD, ETH-USD, and SOL-USD trades and Level-2 book updates, seven days per asset, encoded into $|\mathcal{E}| = 62$ atomic types ($20+20+2+20$ over LO/CO/MO/IS $\times$ side $\times$ level). Genuine test events scored: 3,226,821 (BTC), 3,623,575 (ETH), 2,000,638 (SOL). Trades are attributed to book changes by inter-update timestamp interval and netted against liquidity changes at the traded price; residual decreases are cancellations. Multiple atomic events from one update share its timestamp and are ordered deterministically (MO, CO, LO, IS; bid before ask; then by level). Order sizes are retained in the dataset but unused by the models in this paper.